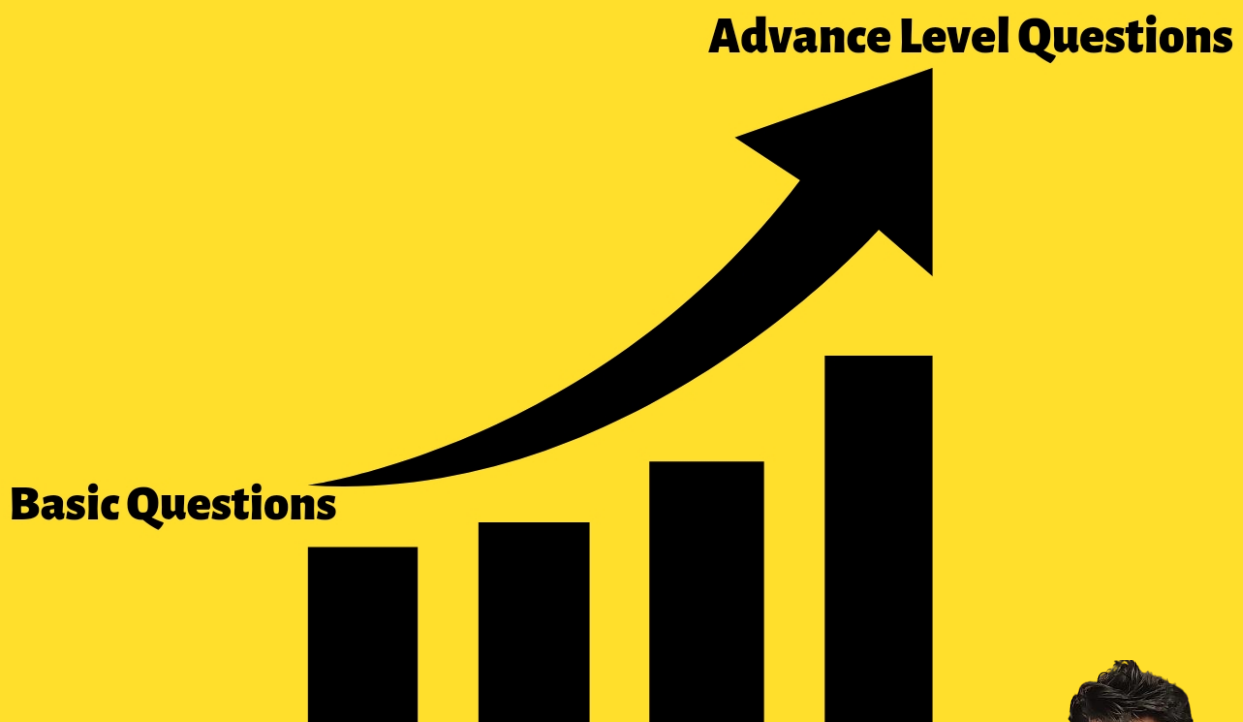


MATHS

PERCENTAGE

ALL TYPE WISE QUESTIONS



Maths with Gajanand



Percentage (प्रतिशत) %

Fraction

1	$= 100\%$	$\frac{1}{16}$	$= 6\frac{1}{4}\%$
$\frac{1}{2}$	$= 50\%$	$\frac{1}{17}$	$= 5\frac{15}{17}\%$
$\frac{1}{3}$	$= 33\frac{1}{3}\%$	$\frac{1}{18}$	$= 5\frac{5}{9}\%$
$\frac{1}{4}$	$= 25\%$	$\frac{1}{19}$	$= 5\frac{5}{19}\%$
$\frac{1}{5}$	$= 20\%$	$\frac{1}{20}$	$= 5\%$
$\frac{1}{6}$	$= 16\frac{2}{3}\%$	$\frac{1}{25}$	$= 4\%$
$\frac{1}{7}$	$= 14\frac{2}{7}\%$	$\frac{1}{40}$	$= 2\frac{1}{2}\%$
$\frac{1}{8}$	$= 12\frac{1}{2}\%$		
$\frac{1}{9}$	$= 11\frac{1}{9}\%$		

Some other Important Fraction

$\frac{1}{10}$	$= 10\%$	$\frac{2}{3}$	$= 66\frac{2}{3}\%$	$\frac{7}{8}$	$= 87.5\%$
$\frac{1}{11}$	$= 9\frac{1}{11}\%$	$\frac{5}{6}$	$= 83\frac{1}{3}\%$	$\frac{4}{11}$	$= 36.36\%$
$\frac{1}{12}$	$= 8\frac{1}{3}\%$	$\frac{4}{5}$	$= 80\%$	$\frac{5}{7}$	$= 71\frac{3}{7}\%$
$\frac{1}{13}$	$= 7\frac{9}{13}\%$	$\frac{3}{4}$	$= 75\%$	$\frac{4}{9}$	$= 44.44\%$
$\frac{1}{14}$	$= 7\frac{1}{7}\%$	$\frac{3}{8}$	$= 37.5\%$	$\frac{7}{9}$	$= 77\frac{7}{9}\%$
$\frac{1}{15}$	$= 6\frac{2}{3}\%$	$\frac{5}{8}$	$= 62.5\%$		

$$\% = \frac{1}{100}$$

Que 1 What is 20% of 500

Soln:-

$$500 \times \frac{20}{100} = \boxed{100} \text{ Ans}$$

2 What is 84% of 12 km.

Soln

$$12 \times \frac{84}{100} = \boxed{10.08} \text{ Ans}$$

3 65g is what percent of 2 kg.?

Soln

$$\frac{65g}{2000g} \times 100 = \boxed{3.25\%} \text{ Ans}$$

4 The time duration of 1 hour 45 minute is what Percent of a day

Soln

$$1 \text{ hour} + \frac{45}{60} \text{ hour} = 1.75 \text{ hour}$$

$$\% = \frac{1.75}{24} \times 100 = \boxed{7.291\%} \text{ Ans}$$

5 What is the difference between 0.6 and 0.6%

Soln:- $0.6\% = \frac{0.6}{100} = 0.006$

$$0.6 - 0.006 = \boxed{0.594} \text{ Ans}$$

Q.6 0.08% of 120% of 50,000 is equal to ?

Soln :- $50,000 \times \frac{0.08}{100} \times \frac{120}{100} \Rightarrow \boxed{48} \text{ Ans}$

7 If x% of a is the same as y% of b, then z% of b will be ?

Soln $x\% \text{ of } a = y\% \text{ of } b$

$$\frac{x}{100} \times a = \frac{y}{100} \times b$$

$$\frac{x}{y} \times a = b \Rightarrow z\% \text{ of } b = \boxed{\frac{xz}{y} \% \text{ of } a}$$

8 A number which, when increased by 16% becomes 1914. The number is:

Soln **m-I**

let number = x

$$x + \frac{16x}{100} = 1914$$

$$\frac{116x}{100} = 1914$$

$$x = \frac{1914 \times 100}{116}$$

$$= \boxed{1650} \text{ Ans}$$

m-II

let number = 100

Increased 16% = 16

$$116 \longrightarrow 1914$$

$$1 \longrightarrow \frac{1914}{116}$$

$$100 \longrightarrow \frac{1914}{116} \times 100$$

$$= \boxed{1650} \text{ Ans}$$

m-III Fraction method

$$16\% = \frac{4}{25}$$

$$\frac{1914}{25} \times 25 = \boxed{1650} \text{ Ans}$$

[9] When a number is increased by 216, it becomes 140% of itself. what is the number?

Soln

$$\begin{array}{ccc}
 100 & & 140 \\
 & \text{---} & \\
 & + 40\% \longrightarrow & 216 \\
 & 100\% \longrightarrow & \frac{216}{40} \times 100 = \boxed{540} \text{ Ans}
 \end{array}$$

[10] If 85% of a number is added to 75, then the result is the number itself. The number is?

Soln [m-I]

$$\begin{aligned}
 x \times \frac{85}{100} + 75 &= x \\
 75 &= x - \frac{85x}{100} \\
 75 &= \frac{15x}{100} \\
 \boxed{500} &= x
 \end{aligned}$$

[m-II]

$$\begin{array}{ccc}
 85\% & \xrightarrow{+75} & 100\% \\
 & \text{---} & \\
 & 15\% \longrightarrow & 75 \\
 & 100\% \longrightarrow & \frac{75}{15} \times 100 \\
 & & = \boxed{500}
 \end{array}$$

[11] When 75 is added to 75% of a number, the answer is the number. Find 40% of that number

Soln

$$\begin{aligned}
 x \times 75\% + 75 &= x \\
 25x\% &= 75 \\
 x &= \frac{75}{25} \times 100 \\
 x &= \boxed{300} \text{ Ans} \\
 40\% \text{ of } x &= \boxed{120} \text{ Ans}
 \end{aligned}$$

[m-II]

$$\begin{array}{ccc}
 75\% & & 100\% \\
 & \text{---} & \\
 & 25\% \longrightarrow & 75 \\
 & 100\% \longrightarrow & \boxed{300}
 \end{array}$$

$$\begin{aligned}
 40\% \text{ of } 300 \\
 = \boxed{120} \text{ Ans}
 \end{aligned}$$

12 Two numbers are respectively 25% and 60% more than a third number. The ratio of the two numbers is:

Soln:-

I	II	III
125	: 160	100 (let)

$$\boxed{25 : 32} \text{ Ans}$$

13 First and second number are less than a third number by 20% and 40% respectively. The second number is less than the first by ??

Soln

I	II	III
80	60	100



$$\% = \frac{-20}{80} \times 100 = \boxed{25\%} \text{ Ans}$$

14 Two numbers are 30% and 37% lesser than a third number. By how much percent is the second number to be enhanced to make its equal to the first number?

Soln

I	II	III
70	63	100



$$\begin{aligned} \% \uparrow &= \frac{+7}{63} \times 100 \\ &= \frac{1}{9} \times 100 \\ &= \boxed{11.11\%} \text{ Ans} \end{aligned}$$

[15] Two numbers are 90% and 75% lesser than a third number. By what % should the first number be increased so that it becomes equal to the second number?

Soln:-

I	II	III
10	25	100

$$\% \uparrow = \frac{+15}{10} \times 100 = \boxed{150\%} \text{ Ans}$$

[16] If the income of A is 40% more than the income of B, then by what percentage is B's income less than income of A?

Soln

A	B
140	100

$$\frac{40}{140} \times 100 = \boxed{\frac{200}{7}\%}$$

m-II

$$40\% = \frac{2}{5}$$

A	B
7	5

$$\frac{2}{7} \times 100 = \boxed{\frac{200}{7}\%}$$

[17] If A is 28% more than B and C is 25% less than the sum of A and B, then by what percent will C be more than A (Correct to one decimal place)?

Soln

A	B	C
128	100	$228 \times \frac{3}{4} = 171$

$$\frac{+43}{128} \times 100 = \boxed{33.6\%}$$

[SSC CGL MAINS 2019]

18 A, B and C are three student. A got 18% more marks than B and 12% less than C. If B got 220 marks, then how much marks C has got?

Soln:-

A
118

B
100 (let)
↓ Given
220

C
 $\frac{118}{88} \times 100$

$$C \rightarrow \frac{220}{100} \times \frac{118}{88} \times 100$$

$$= \boxed{295} \text{ Ans}$$

m-II Fraction method

$$18\% = \frac{9}{50} \Rightarrow \frac{A}{B} = \frac{59}{50}$$

$$12\% = \frac{3}{25} \Rightarrow \frac{A}{C} = \frac{22}{25}$$

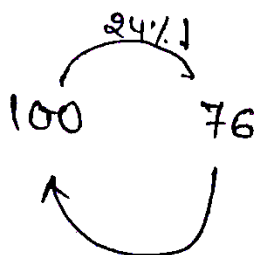
$$\Rightarrow \frac{220}{1100} \times \frac{(C)}{1475}$$

$$= \boxed{295} \text{ Ans}$$

A	:	B	:	C
22×59		50		
59×22				25
1298	:	1100	:	1475
A		B		C

19 The price of an article is cut by 24%, to restore to its original value, the new price must be increased by?

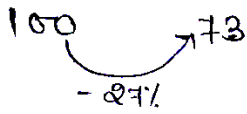
Soln



$$\frac{24}{76} \times 100 = \boxed{31.58\%} \text{ Ans}$$

Q0 If 27% of an electricity bill is deducted, Rs 1825 is still to be paid. How much was the bill?

Soln



$$73 \rightarrow 1825$$

$$100\% \rightarrow \frac{1825}{73} \times 100 = \boxed{2500} \text{ Ans}$$

Q1 A number is increased by $x\%$ to get back to the original number, it is to be reduced by?

Soln:-



$$\frac{-x}{100+x} \times 100 = \frac{100x}{100+x} \% \text{ Ans}$$

Q2 A clerk received an annual salary of ₹ 3,660 in the year 1975. This was 20% more than his salary in 1974. What was his salary in 1974?

Soln

$$20\% = \frac{1}{5}$$

1974

1975

$$5 : 6 \rightarrow 3660$$

$$1 \rightarrow 610$$

$$5 \rightarrow 610 \times 5 = \boxed{3050} \text{ Ans}$$

Q3 The difference between the value of the number increased by 20% and the value of the number decreased by 25% is 36. Find the number.

Soln:-



$$45\% \rightarrow 36$$

$$100\% \rightarrow \frac{36}{45} \times 100 = \boxed{80} \text{ Ans}$$

24] If 25% of 400 + 35% of 1260 + 27% of 1800 = 1020 + x, then the value of x lies between,

- (a) 6 to 10 (b) 0 to 5 (c) 11 to 15 (d) 16 to 20

Soln

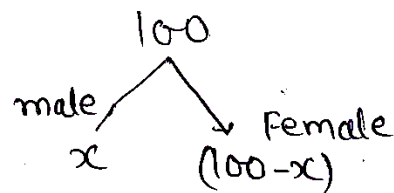
$$\frac{1}{4} \times 400 + \frac{7}{20} \times 1260 + \frac{27}{100} \times 1800 = 1020 + x$$

$$100 + 441 + 486 = 1020 + x$$

$$x = 7 \quad \text{(a) option}$$

25] The sum of the number of male and female students in an institute is 100. If the number of male students is x, then the number of female students becomes x% of the total number of students. Find the number of male students.

Soln:-



$$x\% \text{ of } 100 = 100 - x$$

$$x = 100 - x$$

$$x = 50 \quad \text{Ans}$$

26] The difference of two numbers is 15% of their sum. the ratio of the larger number to the smaller number is:

Soln

$$15\% = \frac{3}{20}$$

$$\frac{A - B}{A + B} = \frac{3}{20}$$

A → large no.
B → small no.

$$20A - 20B = 3A + 3B$$

$$17A = 23B$$

$$\frac{A}{B} = \frac{23}{17} \quad \text{Ans}$$

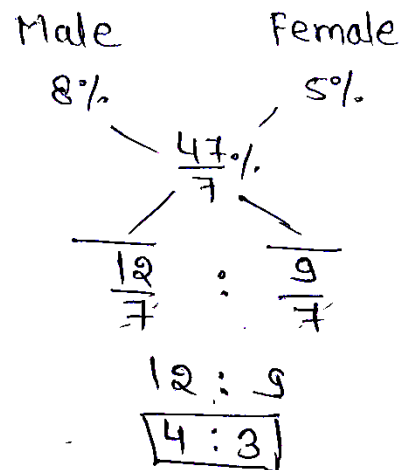
[27] The population of a village was 9800. In a year with the increase in population of males by 8% and that of females by 5%, the population of the village became 10458. What was the number of males in the village before increase?

Solⁿ

$$\begin{array}{ccc} 9800 & & 10458 \\ & \curvearrowright & \\ \text{total } \uparrow & = & \frac{658}{9800} \times 100 = \frac{47}{7}\% \end{array}$$

$$\begin{aligned} \text{So No. of male} &= \frac{9800}{(4+3)} \times 4 \\ &= \boxed{5600} \text{ Ans} \end{aligned}$$

Alligation method



[28] The ratio of the number of boys and girls in a college is 3:2, if 20% of boys and 25% of girls are adults, the percentage of those students who are not adults is:

Solⁿ:-

	Boys	Girls	$\Rightarrow$ Total
let	30	20	50
not adults	$\downarrow 80\%$ 24	$\downarrow 75\%$ 15	
	$\curvearrowright$ +		
	$= \frac{39}{50} \times 100 = \boxed{78\%} \text{ Ans}$		

- [29] In a motor of 120 machine parts, 5% parts were defective. In another motor of 80 machine parts, 10% parts were defective. For the two motors considered together, the percentage of defective machine parts were?

Soln

I	II	Total	
120	80	= 200	
↓ 5%	↓ 10%		% = $\frac{14}{200} \times 100$
6	8	⇒ 14	= $\boxed{7\%}$ <u>Ans</u>

- [30] Kamal has some apples. If he sold 40% more apples more than he ate if he sold 70 apples then. How many he ate

Soln $\left[40\% = \frac{2}{5}\right]$

ate	sold
5	7
×10	↓ ×10
$\boxed{50}$	70
<u>Ans</u>	

- [31] If A is 40% less than B and C is 40% of the sum of A and B, then by what percentage is B greater than C?

Soln

A	B	C
60	100	$(100+60) \times 40\% = 64$
$\frac{36}{64} \times 100 = \boxed{56\frac{1}{4}\%}$ <u>Ans</u>		

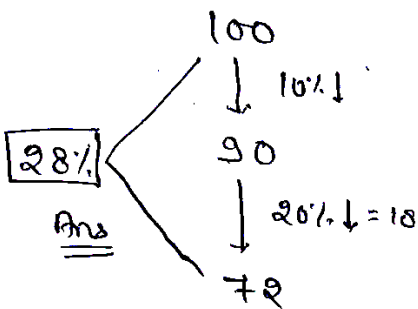
Type - Successive Increase/Decrease

32 A number is decreased by 10% and the resulting number is again decreased by 20%. What is the final percentage of decrease?

Soln

m-I

Basic



m-II

$$10\% = \frac{1}{10}$$

$$20\% = \frac{1}{5}$$

$$\frac{10}{50} \quad \frac{9}{36}$$

$$\frac{14}{50} \times 100 = \boxed{28\%}$$

m-III

$$A \pm B \pm \frac{AB}{100}$$

+ → for increase

- → for Decrease

$$\Rightarrow -10\% - 20\% + \frac{(-10)(-20)}{100}$$

$$\Rightarrow -30\% + 2\%$$

$$\Rightarrow -28\% \text{ Ans } \quad (- \text{ sign shows decrease \%})$$

33 After deducting 60% from a certain number and then deducting 15% from the remainder, 1428 is left. What was the initial number?

Soln

$$A + B + \frac{AB}{100}$$

$$-60 - 15 + \frac{60 \times 15}{100}$$

$$\Rightarrow -66\%$$

rest %

$$\Rightarrow 100 - 66 = 34\%$$

$$\frac{1428}{34} \times 100$$

$$= \boxed{4200} \text{ Ans }$$

m-I let initial number = x

$$60\% = \frac{3}{5}$$

$$15\% = \frac{3}{20}$$

$$x \times \frac{2}{5} \times \frac{17}{20} = 1428$$

$$x = 4200 \quad \underline{\underline{\text{Ans}}}$$

34 The price of an article was increased two times successively by 10% each time. By what percent should the new price be reduced so as to restore the original price.

Solⁿ:- m-I

$$\begin{array}{cc} 10 & 11 \\ 10 & 11 \\ \hline 100 & 121 \end{array}$$

$$\frac{21}{121} \times 100 = 17.36\%$$

m-II

$$\begin{array}{c} 100 \\ \downarrow 10 \\ 110 \\ \downarrow 10 \\ 121 \end{array}$$

$$\frac{21}{121} \times 100 = 17.36\%$$

35 The value of a motorcycle depreciates every year by 4%. what will be its value after 2 years, if its present value is ₹75,000?

Solⁿ m-I $4\% = \frac{1}{25}$

$$\Rightarrow 75000 \times \frac{24}{25} \times \frac{24}{25}$$

$$\Rightarrow \underline{\underline{69120}} \quad \underline{\underline{\text{Ans}}}$$

m-II

$$A + B + \frac{AB}{100} \Rightarrow 2x + \frac{x^2}{100}$$

For when Rate is same

$$\text{overall \%} = 2 \times 4 + \frac{16}{100} = 7.84\%$$

$$75000 \times 108.16\% = \underline{\underline{69120}} \quad \underline{\underline{\text{Ans}}}$$

36 Present population of a village is 67600. It has been increasing annually at the rate of 4%. What was the Population of the village two years ago?

Soln $\left[4\% = \frac{1}{25}\right]$

$$x \times \frac{26}{25} \times \frac{26}{25} = 67600$$

$$\boxed{x = 62500} \text{ Ans}$$

$$(25)^2 : (26)^2$$

$$625 : 676 \rightarrow 67600$$

$$1 \rightarrow 100$$

$$625 \rightarrow \boxed{62500} \text{ Ans}$$

37 The income of a company increases 20% per annum.

If its income is ₹ 26,64,000 in the year 2012.

Then its income in the year 2010 was.

Soln $\left[20\% = \frac{1}{5}\right]$ $x \times \frac{6}{5} \times \frac{6}{5} = 2664000$

$$\boxed{x = 18,50,000}$$

m-II

$$5^2 : 6^2$$

$$25 : 36 \rightarrow 26,64,000$$

$$\Rightarrow \frac{2664,000}{36} \times 25$$

$$\Rightarrow \boxed{18,50,000} \text{ Ans}$$

38

In a factory, the production of cycles rose to 48,400 from 40,000 in 2 years. Then rate of growth per annum is?

Soln

$$40,000 : 48,400$$

$$\sqrt{100} : \sqrt{121}$$

$$10 : 11 \Rightarrow \boxed{10\%} \text{ Ans}$$

m-II

$$\frac{x^2}{40,000} : \frac{y^2}{48,400} \Rightarrow \frac{x^2}{100} : \frac{y^2}{121}$$

$$\Rightarrow \frac{x}{10} : \frac{y}{11}$$

$$\frac{1}{10} \times 100 = \boxed{10\%} \text{ Ans}$$

39 The population of a town has increased by 5% at the end of the first year and decreased by 4% at the end of second year. If the population at end of second year was 55,12,248, then the population at the beginning of first year was:

Soln:-

$$x \times \frac{21}{20} \times \frac{24}{25} = 55,12,248$$

$$\boxed{x = 54,68,500}$$

m-II

$$5\% = \frac{1}{20}, \quad 4\% = \frac{1}{25}$$

$$5\% \uparrow \quad 20 \quad 21$$

$$4\% \downarrow \quad \frac{25}{500} \quad \frac{24}{494}$$

$$= \frac{55,12,248}{494} \times 500$$

$$= \boxed{54,68,500} \text{ Ans}$$

40 A wholesaler sells a watch to a retailer at a profit of 8% and the retailer sells it to a customer at a profit of 12%. If the customer pays Rs. 8,448, what had it cost to the wholesaler?

Soln

$$x \times \frac{27}{25} \times \frac{28}{25} = 8448$$

$$x = \boxed{6981} \text{ Ans}$$

41 A number is first increased by 16% and then increased by 14%. The number, so obtained, is now decreased by 30%. what is the net increase or decrease Percent in the original number (nearest to an integer)?

Solⁿ:- **m-I**

$$\begin{array}{r}
 25 \quad 29 \\
 50 \quad 57 \\
 10 \quad 7 \\
 \hline
 12500 : 11,571 \\
 \quad \quad \quad \ominus \quad \curvearrowright \\
 = \frac{929}{12500} \times 100 \\
 = 7.432\% \\
 \approx \boxed{7\%} \text{ Ans}
 \end{array}$$

m-II

$$\begin{aligned}
 A \pm B \pm \frac{AB}{100} \\
 &= 16 + 14 + \frac{16 \times 14}{100} \\
 &= 32.24\% \quad \& - 30\% \\
 \text{again} \quad &= 32.24 - 30 - \frac{30.24 \times 30}{100} \\
 &= -7\% \quad \text{Ans}
 \end{aligned}$$

m-III

$$x + y + z + \frac{xy}{100} + \frac{yz}{100} + \frac{zx}{100} + \frac{xyz}{10,000}$$

$$x = +16\%$$

$$y = +14\%$$

$$z = -30\%$$

Put & solve we get = $\boxed{-7.432\%}$ Ans

42 The value of a machine is 6250. It decreases by 10% during the first year, 20% during the second year and 30% during the third year. What will be the value of the machine after 3 years?

Solⁿ

$$= 6250 \times \frac{9}{10} \times \frac{4}{5} \times \frac{7}{10}$$

$$= \boxed{3150} \text{ Ans}$$

$$10\% \rightarrow \frac{1}{10}$$

$$20\% \rightarrow \frac{1}{5}$$

$$30\% \rightarrow \frac{3}{10}$$

[43] The strength of a school increases and decreases in every alternate year by 10%. It started with increase in 2000, then the strength of the school in 2003 as compared to that in 2000 was.

Soln

2000 →	10	:	11
2001 →	10		9
2002 →	10		11
2003 →	<u>1000</u>		<u>1089</u>

$$\frac{89}{1000} \times 100 = \boxed{8.9\%} \text{ Inc}$$

[44] A number first increased by 40% and decreased by 25% again increased by 15% and then decreased by 20%. what is the net increase/decrease percent in the number?

Soln

↑	5	:	7
↓	4	:	3
↑	20	:	23
↓	<u>5</u>	:	<u>4</u>
	500		483

$$40\% \Rightarrow \frac{2}{5}$$

$$25\% \Rightarrow \frac{1}{4}$$

$$15\% \Rightarrow \frac{3}{20}$$

$$20\% \Rightarrow \frac{1}{5}$$

$$\downarrow \% = \frac{17}{500} \times 100 = \boxed{3.4\%} \text{ Decrease}$$

[45] The red blood cells in a blood sample grows by 10% per hour in first two hours, decreases by 10% in next one hour, remains constant in next one hour and again increases by 5% per hour in next two hours. If the original count of the red blood cells in the sample is 40,000, find the approximate red blood cell count at the end of 6 hours.

Soln

1 st hour	→	10	11
2 nd hour	→	10	11
3 rd & 4 th hour	→	10	9
5 th hour	→	20	21
6 th hour	→	20	21
		<u>400000</u>	<u>480249</u>

Given → 40,000 : $\boxed{48024.9} \approx \boxed{48025}$ Ans

[46] A businessman's earning by 25% in one year but decreases by 4% in the next. Going by this pattern, after 5 years, his total earnings would be Rs. 72000 what is his present earning?

Soln :-

$$25\% = \frac{1}{4} \uparrow, \quad 4\% = \frac{1}{25} \downarrow$$

$$x \times \frac{5}{4} \times \frac{24}{25} \times \frac{5}{4} \times \frac{24}{25} \times \frac{5}{4} = 72000$$

$$\boxed{x = 40,000}$$

47 If the length of a rectangle is increased by 20% and the breadth is decreased by 10%, the net effect on the area is:

Soln:-

$$\begin{array}{rcl} \text{length :-} & 5 & : 6 \\ \text{breadth :-} & 10 & : 9 \\ \hline \text{Area} & 50 & : 54 \end{array}$$

$$\frac{4}{50} \times 100 = 8\% \uparrow$$

$$\left. \begin{array}{l} 20\% = \frac{1}{5} \\ 10\% = \frac{1}{10} \end{array} \right\}$$

m-II

$$A + B + \frac{AB}{100}$$

$$\Rightarrow +20 - 10 + \frac{(20)(-10)}{100} \Rightarrow 8\% \text{ Ans}$$

48 If each side of a square is decreased by 17%, then by what percentage does its area decrease?

Soln

m-III

$$\begin{array}{rcl} \text{side :} & 100 & : 83 \\ \hline \text{Area} & = (100)^2 & : (83)^2 \\ & = 10000 & : 6889 \end{array}$$

$$\frac{3111}{10000} \times 100 = \underline{\underline{31.11\%}}$$

m-II

$$A + B + \frac{AB}{100}$$

$$-17 - 17 + \frac{17 \times 17}{100}$$

$$\boxed{-31.11\%} \downarrow$$

49 If the radius of a circle is increased by 29%, its area increases by ?

Soln

$$\text{Area} \propto (\text{radius})^2 \quad \text{so} \Rightarrow A + B + \frac{AB}{100}$$

$$\Rightarrow +29 + 29 + \frac{29 \times 29}{100}$$

$$\Rightarrow \boxed{66.41\%}$$

[50] If each edge of a cube is increased by 10% then the percentage increase in its surface area is:

Soln:- $A + B + \frac{AB}{100}$
 $\Rightarrow 10 + 10 + \frac{10 \times 10}{100} \Rightarrow \boxed{21\%}$ Ans

[51] If each edge of a cube is increased by 20% then the percentage increase in its volume is:

Soln $[\text{Volume} \propto (\text{side})^3]$
 $20\% = \frac{1}{5}$
 $(5)^3 : (6)^3$
 Volume: 125 : 216
 $\frac{91}{125} \times 100 = \boxed{72.8\%}$ Ans

[52] The radius of a sphere is increased by 140% By what Percent will its volume increase?

Soln:- $r \rightarrow 10 : 24$
 $V \rightarrow 10^3 : 24^3$
 $\Rightarrow 1000 : 13824$
 $100 : 1382.4$
 $\boxed{1282.4\%}$ Ans

$140\% \rightarrow \frac{14}{10}$

[53] Volume of a sphere is increased by 119.7% when its radius is increased then find % change in curved surface area of sphere?

$$V = \frac{4}{3} \pi R^3$$

$$TSA = 4\pi R^2$$

$$V \rightarrow 100 : 219.7$$

$$1000 : 2197$$

$$r \rightarrow \sqrt[3]{1000} : \sqrt[3]{2197}$$

$$10 : 13$$

$$C.S.A \rightarrow 10^2 : 13^2$$

$$100 : 169$$

$$\boxed{69\%} \text{ Ans.}$$

[54] If the height of a cylinder is increased by 15% and the radius of its base is decreased by 10% then the percentage change in its curved surface area is:

Soln:- $15\% \rightarrow \frac{3}{20}$
 $10\% \rightarrow \frac{1}{10}$

Radius $\rightarrow 10 \quad 9$
 Height $\rightarrow \frac{20}{23}$
 Area $\rightarrow \frac{200}{207}$

m-II $\boxed{A + B + \frac{AB}{100}} \Rightarrow 15 - 10 - \frac{150}{100}$
 $\Rightarrow \boxed{3.5\%} \text{ Ans}$

$$\frac{7}{200} \times 100 = \boxed{3.5\%}$$

[55] If the radius of the cylinder is decreased by 20%, then by how much percent the height must be increased, so that the volume of the cylinder remains same?

Soln:- $V = \pi r^2 h \rightarrow$

$$r = 5^2 : 4^2$$

$$h = 4^2 : 5^2$$

$$\frac{+9}{16} \times 100 = \boxed{56.25\%}$$

[56] In a triangle height is increased by 10% and base is increased by 20%. then find the percentage increase in area of triangle?

Soln

$$\begin{array}{lcl} \text{Base} & \rightarrow & 5 : 6 \\ \text{height} & \rightarrow & \frac{10}{50} : \frac{11}{66} \end{array}$$

$$\frac{+16}{50} \times 100 = \boxed{32\%}$$

[m-II]

$$A + B + \frac{AB}{100}$$

$$10 + 20 + \frac{10 \times 20}{100}$$

$$\boxed{32\%} \text{ Ans}$$

[57] The base of a triangle is increased by 40%. By what percentage should its height be increased so that the area increases by 60%? [CGL pre-2022]

Soln:- 40% $\rightarrow \frac{2}{5}$

60% $\rightarrow \frac{3}{5}$

Area $\rightarrow 5 : 8$

Base $\rightarrow 5 : 7$

Height $\rightarrow \frac{5}{5} : \frac{8}{7}$

$7 : 8$

$$\frac{1}{7} \times 100 = \boxed{14\frac{2}{7}\%}$$

[58] In a triangle base is increased by $66\frac{2}{3}\%$, then by how much % height should be reduced such that area of triangle remains constant?

Soln

$66\frac{2}{3}\% \rightarrow \frac{2}{3}$

Base $\rightarrow 3 : 5$

height $\rightarrow 5 : 3$

$$\frac{-2}{5} \times 100 = \boxed{40\%} \text{ Ans}$$

Type

$$\text{Expenditure} = \text{Price} \times \text{consumption}$$

- [59] The price of oil is increased by 20%. However, its consumption decreased by $8\frac{1}{3}\%$. what is the percentage increase or decrease in the expenditure on it?

Soln $E = P \times C$

$$20\% \rightarrow \frac{1}{5}$$

$$8\frac{1}{3}\% \rightarrow \frac{1}{12}$$

$$\begin{array}{ccc} P & \rightarrow & 5 \quad 6 \\ E & \rightarrow & 12 \quad 11 \\ \hline E & & 60 : 66 \end{array}$$

$$\frac{6}{60} \times 100 = \boxed{10\%} \text{ Ans}$$

- [60] The price of sugar has decreased by 15%. By what Percentage can a person increase the consumption so that there is no change in the expenditure?

Soln:- $15\% \rightarrow \frac{3}{20}$

$$P \rightarrow 20 : 17$$

$$C \rightarrow 17 : 20 \quad (\text{Reversed})$$

$$+\frac{3}{17} \times 100 = \boxed{\frac{300}{17}\%} \text{ Ans}$$

- [61] The price of sugar is increased by 20%. If the expenditure on sugar has to be kept the same as earlier, the ratio between the reduction in consumption and the original consumption is:

Soln $20\% \rightarrow \frac{1}{5}$

$$P \rightarrow \begin{array}{cc} \text{Initial} & \text{Final} \\ 5 & 6 \end{array}$$

$$C \rightarrow 6 : 5$$

$$\boxed{1 : 6} \text{ Ans}$$

[62] The price of diesel is increased by 26%. A person wants to increase his expenditure by 15% only. By what percentage, correct to one decimal place, should he decrease his consumption? [CPO-2020]

Soln:-

$$26\% \rightarrow \frac{13}{50}$$

$$15\% \rightarrow \frac{3}{20}$$

$$\begin{array}{l} E \rightarrow 20 : 23 \\ P \rightarrow 50 : 63 \\ \hline C \rightarrow \frac{E}{P} \rightarrow \frac{20}{50} : \frac{23}{63} \end{array}$$

$$126 : 115$$

$$\frac{11}{126} \times 100 = \boxed{8.7\%} \text{ Ans}$$

[63] When the price of sugar decreased by 10%, a man could buy 1kg more for ₹ 270. Then the original price of sugar per kg is :

Soln:-

original Present
[m-1] $P \rightarrow 10 : 9$

$$C \rightarrow 9 : 10$$

$$1 \text{ Unit} \rightarrow 1 \text{ kg}$$

So original Consumption = 9kg

$$\text{Price} = \frac{270}{9}$$

$$= \boxed{30 \text{ ₹/kg}}$$

[m-0]

$$270 \times 10\%$$

$$\Rightarrow 27 \text{ ₹} \rightarrow 1 \text{ kg}$$

$$\text{Present price} = 27 \text{ ₹/kg}$$

$$\text{original price} = \frac{27}{90\%} \times 100\%$$

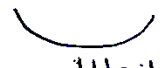
$$= \boxed{30 \text{ ₹/kg}}$$

Ans

[64] If the price of pen decreases by 20% then a man can buy 10 more pens for Rs 100. What is the new Price of each pen?

Soln:- m-1

	old	:	new
P →	5	:	6
C →	6	:	5



 1 Unit → 10 pen

new consumption → 5×10
= 50

Price = $\frac{100}{50} = \boxed{2 \text{ ₹/pen}}$

m-II

$$100 \times 20\% = 20 \text{ ₹}$$

$$20 \text{ ₹} \longrightarrow 10 \text{ pen}$$

$$\boxed{2 \text{ ₹}} \longleftarrow 1 \text{ pen}$$

Ans

[65] If the price of eraser is reduced by 25%. A person can buy 2 more erasers for a rupee. How many erasers are available for a rupee after reduction?

Soln:-

$$P \rightarrow 4 : 3$$

$$C \rightarrow 3 : \textcircled{4}$$



 1 Unit → 2

$$4 \text{ Unit} \rightarrow 4 \times 2$$

$$= \boxed{8 \text{ eraser}} \quad \underline{\underline{\text{Ans}}}$$

66 A reduction in the price of apples enables a person to purchase 3 apples for ₹1 instead of ₹1.25. What is the % of reduction in price? [SSC CGL 2013]

Soln:-

$$\text{Price } 1.25 : 1$$

$$= \frac{-0.25}{1.25} \times 100 = \boxed{20\%} \text{ Ans}$$

67 Price of petrol increased from Rs 60/litre to Rs 75/litre. How much should the consumption of petrol be reduced (in%) so as to increase expenditure by only 10%?

Soln:-

m-I $60\text{ ₹/lit} : 75\text{ ₹/lit}$

$$\text{Increase \%} = \frac{15}{60} \times 100$$

$$\text{Price } \uparrow = 25\%$$

$$\begin{array}{l} E \rightarrow 10 : 11 \\ P \rightarrow 4 : 5 \end{array}$$

$$\frac{C \rightarrow E}{P} \rightarrow \frac{10}{4} : \frac{11}{5}$$

$$50 : 44$$

$$= \frac{6}{50} \times 100 = \boxed{12\%}$$

m-II

$$\text{Price Increase \%} = 25\%$$

$$\begin{array}{c} 100 \\ \swarrow +10\% \quad \searrow +25\% \\ 110 \quad \quad 125 \end{array}$$

$$= \frac{15}{125} \times 100$$

$$= \boxed{12\%} \text{ Ans}$$

68

Type

The price of an article is reduced by 25% but the daily sale of the article is increased by 30%. The net effect on the daily receipt is ?

Soln:-

$$\begin{array}{rcl} \text{Price} & \rightarrow & 4 : 3 \\ \text{Sale} & \rightarrow & \frac{10}{40} : \frac{13}{39} \end{array}$$

$$25\% \rightarrow \frac{1}{4}$$

$$30\% \rightarrow \frac{3}{10}$$

$$\frac{-1}{40} \times 100 = \boxed{2\frac{1}{2}\% \text{ decrease}}$$

69

When the price of an item was reduced by 20%, then its sale increased by $x\%$. If there is an increase of 60% in the receipt of the revenue, then the value of x is ?

Soln:-

$$\text{Revenue :- } 5 : 8$$

$$\text{Price :- } 5 : 4$$

$$\text{Sale} = \frac{R}{P} \rightarrow \frac{5}{5} : \frac{8}{4}$$

$$= 1 : 2$$

$$\boxed{100\%}$$

Ans

Type

- 70** Renu saves 20% of her income. If her expenditure increase by 20% and income increase by 20% then her saving increase by ?

Soln:-

	Income	Exp.	Saving
	100	80	20
	↓	↓	↓
New →	120	96	33
			$\frac{11}{20} \times 100$
			$= \boxed{55\%}$ <u>Ans</u>

- 71** Rishu saves $x\%$ of her income. If her income increases by 26% and the expenditure increases by 20%. then her saving increase by 50%. what is the value of x ?

Soln:-

	Income	Saving	Exp
	100	x	$100 - x$
	↓ 26%	↓ 50%	↓ 20%
New →	126	$1.5x$	$120 - 1.2x$

$$\Rightarrow 126 - 1.5x = 120 - 1.2x$$

$$6 = 0.3x$$

$$\boxed{x = 20} \text{ Ans}$$

72 A man spends 75% of his income. If his income increases by 28% and his expenditure increases by 20% then what is the increase or decrease percentage in his savings?

Soln:-

	I	E	S	
let $\rightarrow$	400	300	100	$75\% = \frac{3}{4}$
	$\downarrow +28\%$	$\downarrow +20\%$	$\downarrow$	
change $\rightarrow$	112	- 60	=	<u><u>52%</u></u> Ans

73 The monthly income of a person was ₹ 13,500 and his monthly expenditure was ₹ 9,000. Next year his income increased by 14% and his expenditure increased by 7%. The percentage increased in his saving was?

Soln:-

I	E	S
13500	: 9000	: 4500
3	: 2	: 1

let the

	Income	Expenditure	Saving
	300	200	100
	$\downarrow +14\%$	$\downarrow +7\%$	$\downarrow \rightarrow +28$
new $\rightarrow$	342	- 214	= 128

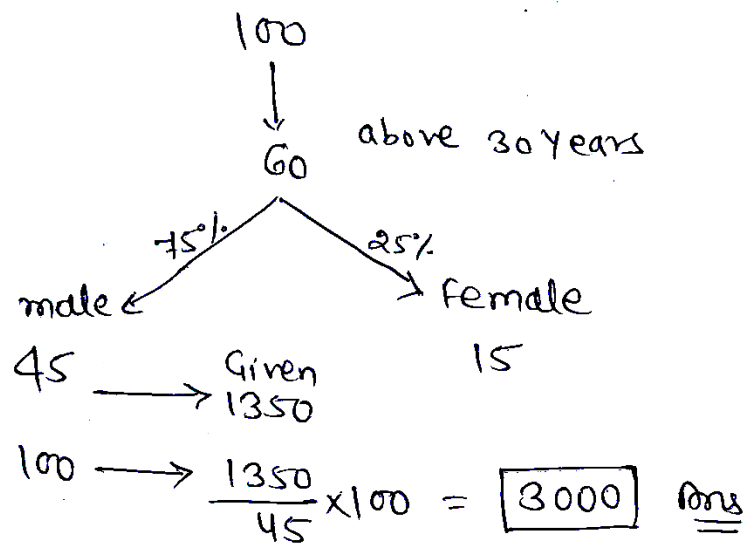
$$\% \text{ chang} = \frac{28}{100} \times 100 = \boxed{28\%} \text{ Ans}$$

74

Type

In a factory 60% of the workers are above 30 years and of these 75% are males and the rest are females. If there are 1350 male workers above 30 years, the total number of workers in the factory is ?

Soln:- let the total numbers of workers = 100



75 The population of a town is 3,11,250. The ratio of woman to men is 43:40. If there are 24% literate among women and 10% illiterate, among men, the total number of literate persons in the town is:

Soln:-

$$\text{Total} = 3,11,250$$

$$\text{woman : men} = 43 : 40$$

$$\text{woman} = \frac{3,11,250}{43+40} \times 43 = 161250$$

$$\text{man} = \frac{3,11,250}{43+40} \times 40 = 150000$$

$$\text{Literate woman} = 161250 \times \frac{24}{100} = 38700$$

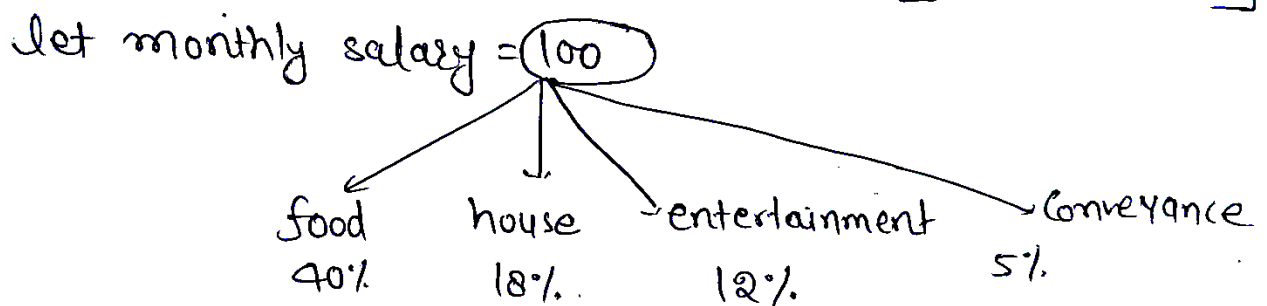
$$\text{Literate man} = 150000 \times \frac{(100-10)}{100} = 135000$$

$$\text{Total literate person} = 38700 + 135000 = 173700$$

Ans

76 Ramesh spends 40% of his monthly salary on food, 18% on house rent, 12% on entertainment, and 5% on conveyance. But due to a family function, he had to borrow Rs. 16,000 from a money lender to meet the expenses of Rs. 20,000. His monthly salary is? [SSC CPO-2020]

Soln.:-



$$\text{Total} = 75\%$$

$$\text{Remaining} = 100 - 75 = 25\%$$

$$\text{Given } 25\% \longrightarrow (20,000 - 16,000)$$

$$100\% \longrightarrow \frac{4000}{25} \times 100$$

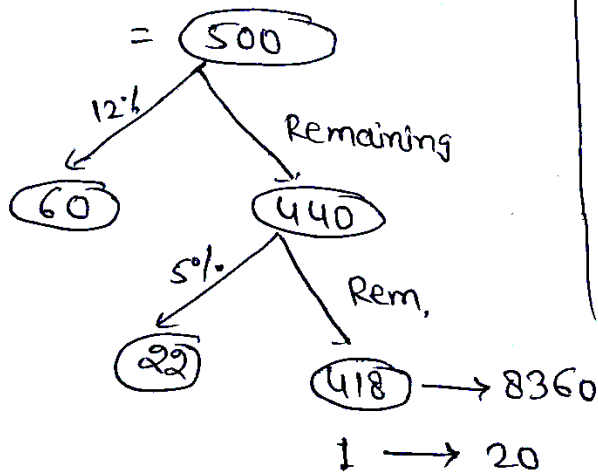
$$= \boxed{16000} \text{ Ans}$$

77 A village lost 12% of its goats in a flood and 5% of remainder died from diseases. If the number left now is 8360. what was the original number before the flood?

Soln:- **m-I** $12\% = \frac{3}{25}$

$$5\% = \frac{1}{20}$$

let goats = 20×25



So $500 \rightarrow 20 \times 500$
 $= \boxed{10,000}$
Ans,

m-II let goats = x

$$x \times \frac{22}{25} \times \frac{19}{20} = 8360$$

$$x = \frac{8360}{\frac{22}{25} \times \frac{19}{20}}$$

$x = \boxed{10,000}$ Ans

m-III $A + B + \frac{AB}{100}$

$$-12 - 5 + \frac{60}{100}$$

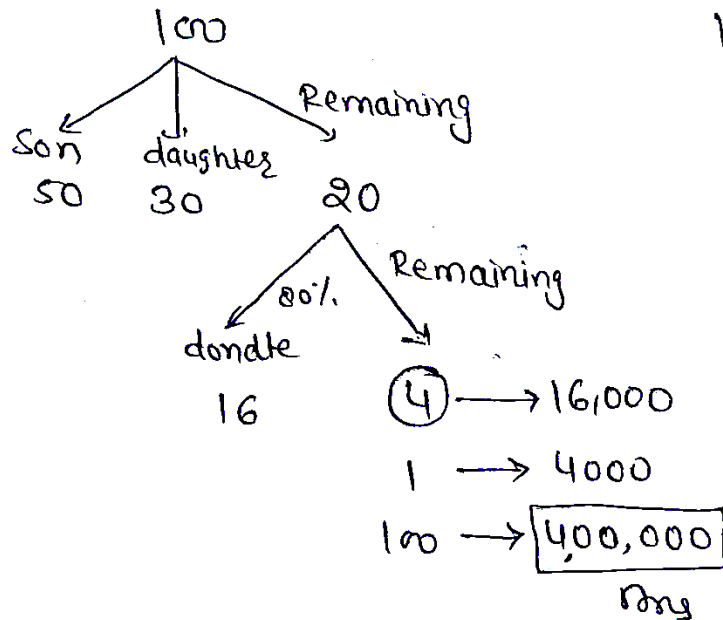
$$\Rightarrow -16.4\%$$

Remaining = $100 - 16.4 = 83.6\% \rightarrow 8360$

$100\% \rightarrow \boxed{10,000}$ Ans

78 A man gives 50% of his money to his son and 30% to his daughter. 80% of the rest is donated to a trust. If he is left with 16,000 now, how much money did he have in the beginning?

Soln



m-II

$$50\% + 30\% = 80\%$$

$$x \times \frac{1}{5} \times \frac{1}{5} = 16000$$

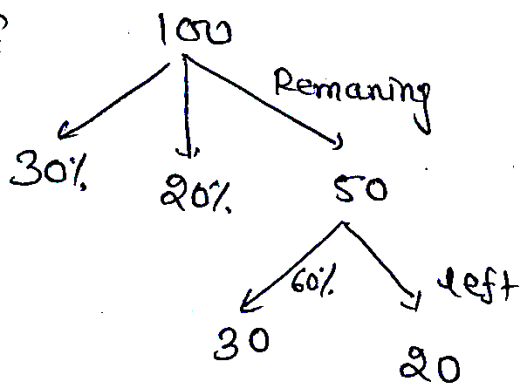
$$x = 16000 \times 5 \times 5$$

$$= \boxed{4,00,000}$$

Ans

79 A person spends 30% of his income on food 20% children's education and 60% of remaining on house rent. What percent of income is left with him?

Soln



% left = $\boxed{20\%}$ Ans

m-II

$$30\% + 20\% = 50\%$$

$$60\% \rightarrow \frac{3}{5}$$

$$x \times \frac{1}{2} \times \frac{3}{5} \leftarrow \text{left}$$

$$\text{Income } x \xrightarrow{\frac{1}{5}} \frac{1}{5} \rightarrow \boxed{20\%}$$

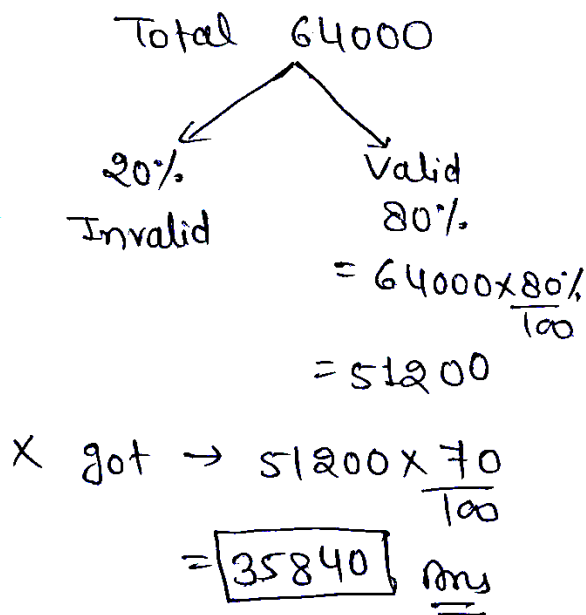
Ans

Type - Election

80 In an election, candidate X got 70% of the overall valid votes. If 20% of the overall votes were declared invalid and the total numbers of votes is 64000, then find the number of valid polled in favour of the candidate.

[SSC CHSL-2020]

Soln:-



m-II $20\% \rightarrow \frac{1}{5}$, $70\% = \frac{7}{10}$

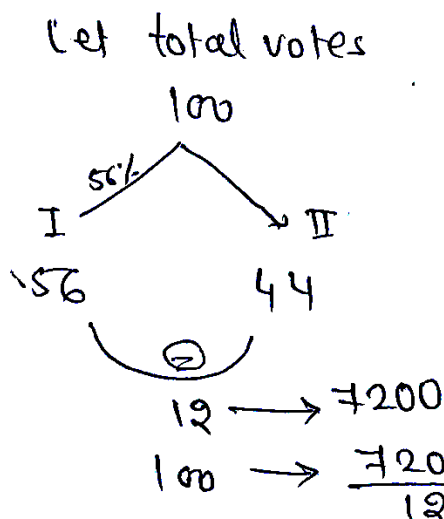
$$64000 \times \frac{4}{5} \times \frac{7}{10}$$

Valid

$$= \boxed{35840} \text{ Ans}$$

81 In an election a candidate who gets 56% of the votes is elected by a majority of 7200 votes. what is total number of voters enrolled on the voter list?

Soln:-



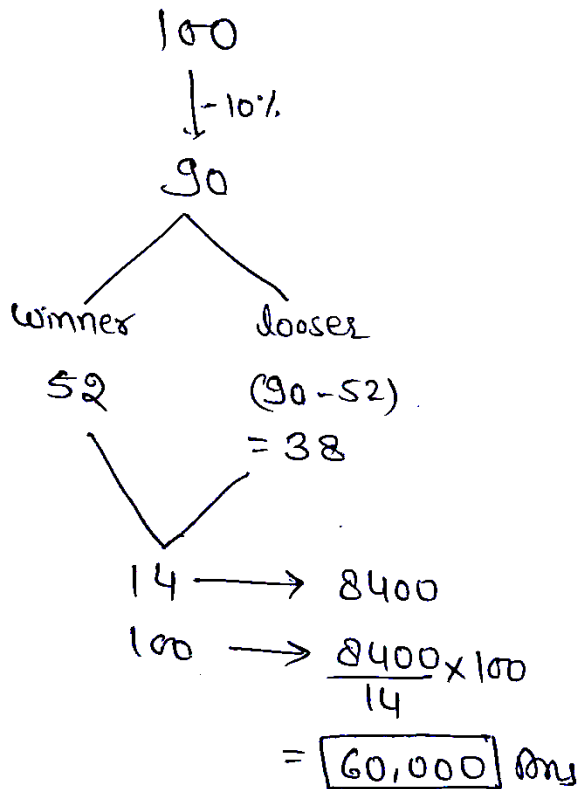
m-II $100 - 56 = 44\%$
 $56\% - 44\% = 12\%$

$$\Rightarrow \frac{7200}{12\%} \times 100\%$$

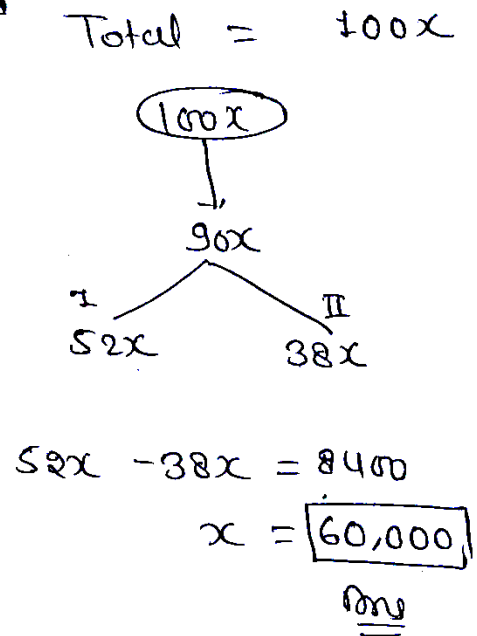
$$\Rightarrow \boxed{60,000} \text{ Ans}$$

82 10% of the voters did not cast their votes in an election between two candidate. The successful candidate got 52% of the total votes and won by 8400 votes. Find the number of votes polled.

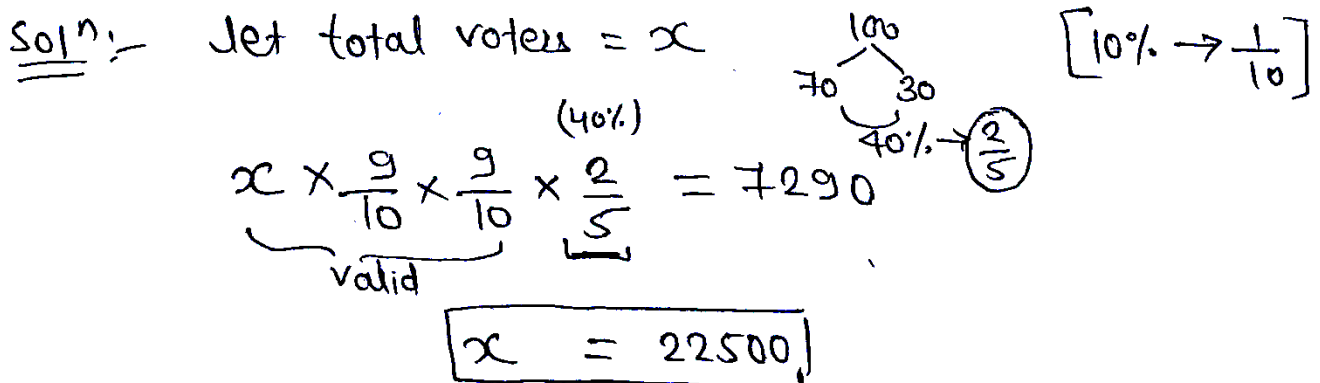
Soln:-



m-II

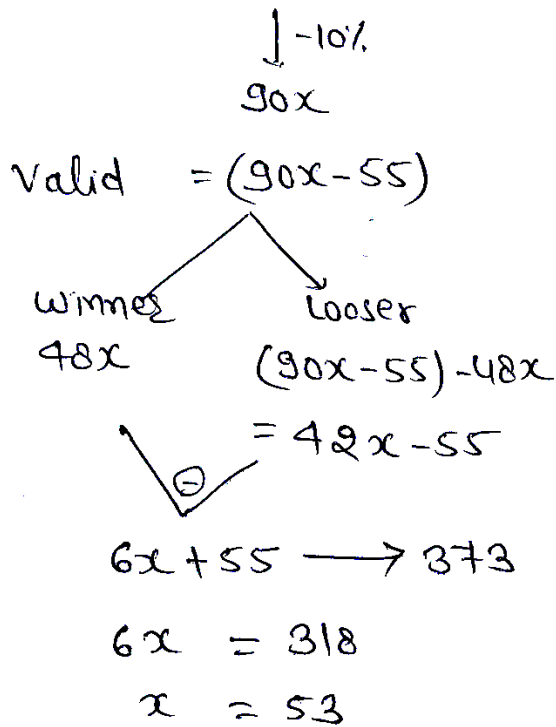


83 In an election two candidate have participated. 10% of total voters did not cast. their votes of the polled votes, 10% are declared as invalid. If the winner got 70% of valid votes, and won by 7290 votes, then find, how many voters are there in the voting list?

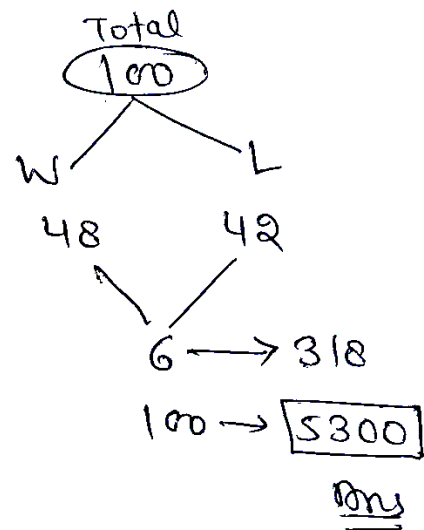


84 10% of the voters did not cast their votes in an election between two candidates. 55 of the votes polled were found invalid the successful candidate got 48% of the voter list and won by a majority of 373 votes. The number of voters enrolled on the voter's list was?

Solⁿ:- ~~Ans~~ total = $100x$



$$373 - 55 = 318$$

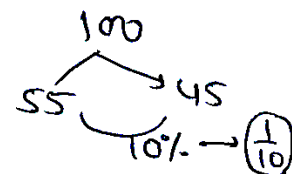


So total vote = $100x$
= 5300 Ans

85 In an election two candidate participated, 10% voters did not vote, 2500 votes declared invalid and the winner gets 55% of the valid votes and win 2000 votes. Find the number of voters in the voting list?

Solⁿ:-

Total x



$$\left[x \times \frac{9}{10} - 2500 \right] \times \frac{1}{10} = 2000$$

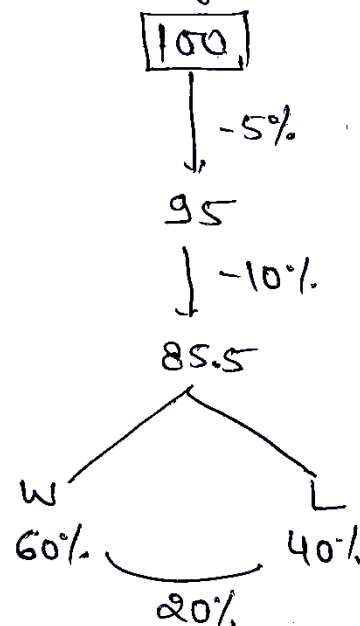
$x = 25000$ Ans

86 In an election between two candidates, 5% of the registered voters did not cast their vote. 10% of the votes were found to be either invalid or of NOTA. The winning candidate received 60% votes in his favour and won the election by 17271 votes. Find the number of registered voters.

[SSC CGL Pre-2022]

Soln:-

m-I Let total registered voters



$$85.5 \times 20\% \rightarrow 17271$$

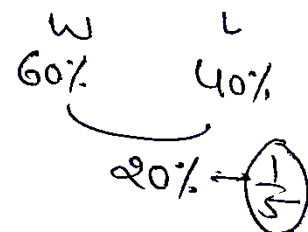
$$100\% \rightarrow \frac{17271}{85.5 \times 20} \times 100 = 101000$$

m-II

$$5\% \rightarrow \frac{1}{20}, \quad 10\% \rightarrow \frac{1}{10}$$

$$\left(x \times \frac{19}{20} \times \frac{9}{10}\right) \times \frac{1}{5} = 17271$$

$$x = \boxed{101000} \text{ Ans}$$



Type

87

Pranita got 30 marks more in math than what she got in science. Her Math marks are 60% of the sum of her Math and science marks. what are her Science marks ?

Soln:-

math	science
$x+30$	x

$$\text{and } x+30 = (x+30+x) \times \frac{60}{100}$$

$$\Rightarrow x+30 = (2x+30) \times \frac{3}{5}$$

$$\Rightarrow 5x+150 = 6x+90$$

$$\boxed{x = 60} \quad \underline{\underline{\text{Ans}}}$$

88 In an examination there are three subjects of 100 marks each. A student scores 60% in the first subject and 80% in the second subject. He scored 70% in aggregate. His percentage of marks in the third subject is ?

Soln:-

$$\begin{aligned} \text{Total marks} &= 100+100+100 \\ &= 300 \end{aligned}$$

$$\begin{aligned} \text{He got } 70\% &= 300 \times \frac{70}{100} \\ &= 210 \end{aligned}$$

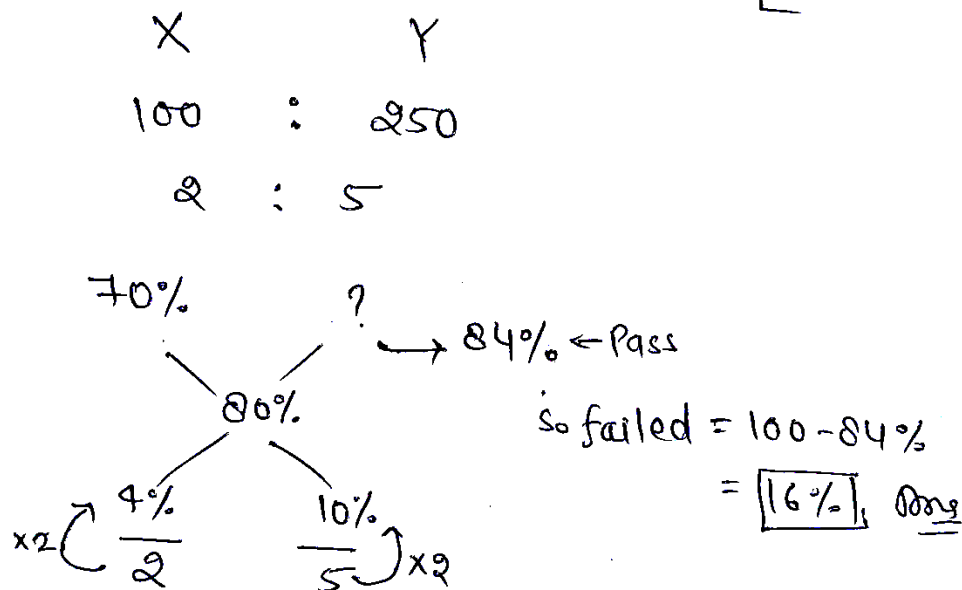
I st	II nd	III rd	
60	+ 80	+ x	= 210

$$\boxed{x = 70} \quad \underline{\underline{\text{Ans}}}$$

- [89] A certain number of student from school X appeared in an examination and 30% student failed. 150% more students than more from school X, appeared in the same examination from school Y, If 80% of the total number of students who appeared from X and Y passed, then what is the percentage of student who failed from Y?

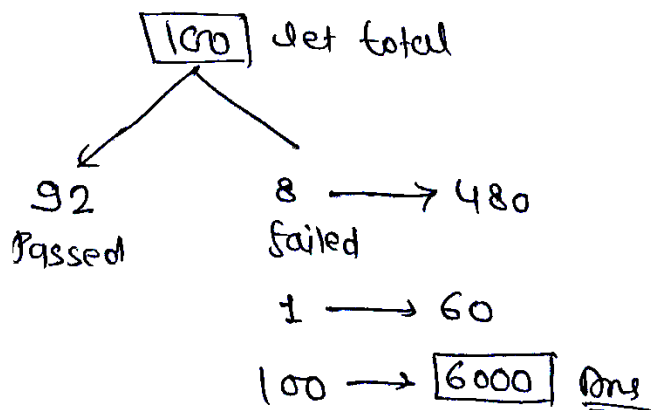
Soln:-

[SSC CGL main 2020]



- [90] In an examination 92% of the students passed and 480 students failed. If so, how many students appeared in the examination?

Soln



91 40% are the passing marks. A student gets 250 marks yet fails by 38 marks. What is the maximum marks?

Soln :- **m-I** - Basic

Let maximum marks = x

$$x \times \frac{40}{100} = 250 + 38$$

$$x = \frac{288}{40} \times 100$$

$$x = \boxed{720} \text{ Ans}$$

m-II

$$250 + 38 = 288$$

$$\frac{288}{40\%} \times 100\%$$

$$= \boxed{720} \text{ Ans}$$

92 In a test a student got 30% marks and failed by 25 marks. In the same test another student got 40% marks and secured 25% marks and more than the essential minimum pass marks. The maximum pass marks for the test were?

Soln

m-I Let Total marks
 $\boxed{100x}$

$$\underbrace{30x + 25}_{\text{Passing marks}} = \text{I}^{\text{st}} \text{ student}$$

$$\underbrace{40x - \frac{(30x + 25) \times 25}{100}}_{\text{Pass marks}} = \text{II}^{\text{nd}} \text{ student}$$

$$\text{So } 30x + 25 = 40x - \frac{(30x + 25) \times 25}{100}$$

$$x = 12.50 \text{ } \nearrow \text{ put}$$

$$\text{Pass mark} = 30x + 25 \Rightarrow 375 + 25 = \boxed{400} \text{ Ans}$$

Type - Tax

93 If the income tax is increased by 19%, the net income is reduced by 1%. the rate of income Tax is :

Soln

$$\text{Tax} \times \frac{19}{100} = \text{N.I} \times \frac{1}{100}$$

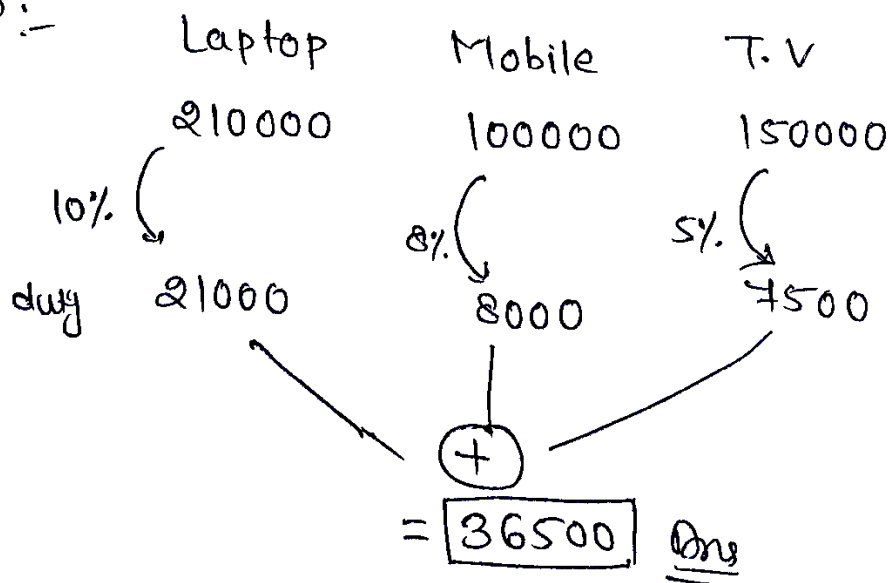
$$\frac{\text{Tax}}{\text{N.I}} = \frac{1}{19} \Rightarrow \text{Total Income}$$

$$\% \text{ rate} = \frac{1}{20} \times 100$$

$$= \boxed{5\%} \quad \underline{\text{Ans}}$$

94 A businessman imported Laptop, worth ₹210000, Mobile phones worth ₹100000 and Television sets worth ₹150000. He had to pay 10% duty on Laptops, 8% on Phones and 5% on Television sets as special case. How much total duty (in Rupees) he had to pay on all items as per above details?

Soln:-



95 A man's annual income has increased by Rs. 1.2 Lakhs but the tax on income that he has to pay has reduced from 12% to 10%. He now pays the same amount of tax as before. what is his increased income (in Rs. Lakhs) ?

Soln:- Let income = x

$$12\% \text{ of } x = (x + 1.2 \text{ Lakh}) \times 10\%$$

$$12x = 10x + 12 \text{ Lakh}$$

$$x = 6 \text{ Lakh}$$

$$\text{Increased income} = (6 + 1.2) = \boxed{7.2 \text{ Lakh}} \text{ Ans}$$

96 A man's annual income has increased by Rs. 5 lakhs but the tax on income that he has to pay has reduced from 12% to 10%. He now pays Rs. 10,000 more income tax. what is his increased income (in Rs. lakhs)?

Soln:- Let income = x lakh

$$(x+5) \times \frac{10}{100} - \frac{12x}{100} = 10,000$$

$$10(x+5) - 12x = 10 \text{ Lakh}$$

$$50 \text{ Lakh} - 2x = 10 \text{ Lakh}$$

$$40 \text{ Lakh} = 2x$$

$$x = 20 \text{ Lakh}$$

$$\text{increased income} = (20 + 5) = \boxed{25 \text{ Lakh}} \text{ Ans}$$

[97] A student multiplied a number with $\frac{3}{4}$ instead of $\frac{4}{3}$, what is error percentage?

Soln:- wrong $\frac{3}{4}$ $\times$ $\frac{4}{3}$ correct

9 : 16

$\frac{7}{16} \times 100 = 43 \frac{3}{4} \%$

[98] A line of length 1.5 metres was measured as 1.55 metres by mistakes, what will be the value of error percentage?

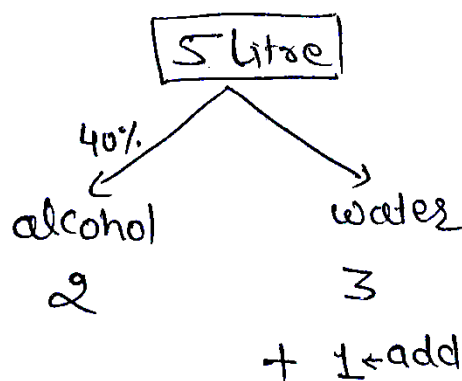
Soln:-

$\boxed{\checkmark}$ 1.5 $\quad$ $\boxed{\times}$ 1.55

$\frac{0.05}{1.5} \times 100 = 3 \frac{1}{3} \%$ Ans

[99] 1 litre of water is added to 5 litres of alcohol water solution. Containing 40% alcohol strength. The strength of alcohol in the new solution will be?

Soln



alcohol % = $\frac{2}{(5+1)} \times 100$

= $\frac{2}{6} \times 100$

= $33 \frac{1}{3} \%$ Ans

100 75 gm of sugar solution has 30% sugar in it. Then the quantity of sugar that should be added to the solution to make the quantity of the sugar 70% in the solution is ?

Soln:-

75 gm

Sugar

$$75 \times 30\% + x = (75 + x) \times 70\%$$

$$x = (75 + x) \times \frac{7}{10} - 75 \times \frac{30}{100}$$

$$x - \frac{7x}{10} = 52.5 - 22.5$$

$$\frac{3x}{10} = 30$$

$$x = \boxed{100 \text{ gm}} \text{ Ans}$$

101 If the numerator of a fraction is increased by 60% and the denominator is increased by 40%, then resultant fraction is $\frac{16}{23}$, The original fraction is ?

Soln $60\% = \frac{3}{5}$

$40\% = \frac{2}{5}$

let numerator = $5x$

denominator = $5y$

$$\frac{5x + 3x}{5y + 2y} = \frac{16}{23} \Rightarrow \frac{x}{y} = \frac{2}{9}$$

$$\Rightarrow \frac{5x}{5y} = \boxed{\frac{2}{9}} \text{ Ans}$$

102 Sachin scored 120 runs, which included 6 boundaries and 4 sixes. What percentage of total score did he make by running between the wickets?

Soln:-

$$120 - 6 \times 4 - 4 \times 6$$

$$= 120 - 24 - 24$$

$$= 72$$

$$\% = \frac{72}{120} \times 100 = \boxed{60\%} \text{ Ans}$$

103 What % of a day is 30 minutes?

Soln

$$\frac{30}{24 \times 60} \times 100 = \boxed{2.083\%} \text{ Ans}$$

104 A box has 100 blue, 50 red balls, 50 black balls, 25% of blue balls and 50% of red balls are taken away. Percentage of black balls of present is?

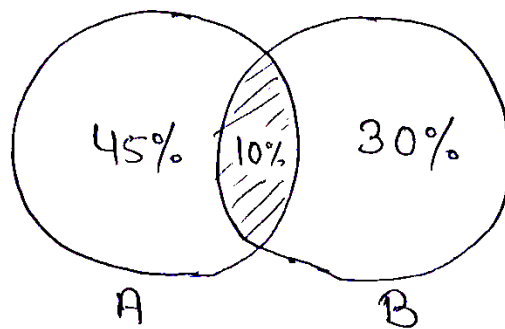
Soln

Blue	Red	Black	Total
100	50	50	$\Rightarrow 200$
$\downarrow 25\%$	$\downarrow 50\%$		
-25	-25		
Remaining 75	25	50	$\Rightarrow 150$

$$\text{Black \%} = \frac{50}{150} \times 100$$

$$= \boxed{33\frac{1}{3}\%} \text{ Ans}$$

Type – Ven Diagram



$$\text{Total} = 100\%$$

$$A = 55\%$$

$$B = 40\%$$

$$A \cap B = 10\%$$

$$\text{only } A = 45\%$$

$$\text{only } B = 30\%$$

$$A \cup B = 45\% + 10\% + 30\% \\ = 85\%$$

$$* \boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)} *$$

105 In an examination, 52% students failed in Hindi and 42% in English. If 17% failed in both the subjects, what percentage of students passed in both the subjects?

Soln :- Total % of failed student = $n(A \cup B)$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$= 52\% + 42\% - 17\%$$

$$= 77\%$$

$$\text{Passed \%} = 100 - 77 = \boxed{23\%} \text{ Ans}$$

106 In an examination, 53% of the candidates failed in science and 48% failed in mathematics. If 40% failed in both subjects, what percentage passed in both subjects?

Soln Failed % = $53 + 48 - 40 = 61\%$

Pass % = $100 - 61 = \boxed{39\%}$ Ans

107 In an examination, 48% of candidates passed in science and 56% failed in mathematics. If 32% failed in both subjects, then what percent passed in both subjects?

Soln:- Science failed % = $100 - 48 = 52\%$

math failed % = 56%

Total failed % = $52 + 56 - 32$
 $= 76\%$

So Pass % = $100 - 76 = \boxed{24\%}$ Ans

108 In an examination 70% of the candidates passed in English. 80% passed in Mathematics. 10% failed in both the subjects. If 144 candidates passed in both the total number of candidates were?

Soln:- Pass in both = $70\% + 80\% - 90\%$ $\swarrow (100-10\%)$
 $= 60\%$

$60\% \longrightarrow 144$

$100\% \longrightarrow \frac{144}{60} \times 100 = \boxed{240}$ Ans

109 Daily local train ticket costs Rs 110 and monthly Pass costs Rs 2524.5. If Ram buys the monthly pass and travel for 27 days in a month than he will save-

Soln:-

Total Cost

$$110 \times 27 = 2970$$

monthly pass cost

$$2524.5$$

$$\frac{445.50}{2970} \times 100 = \boxed{15\%} \text{ Ans}$$

110 There is a 4% increase in volume when a liquid freezes to its solid state. The percentage decrease when solid melts to liquid again, is ?

Soln:- 4% $\rightarrow \frac{1}{25}$

$$\begin{array}{ccc} 25 & & 26 \\ & \nwarrow & \nearrow \\ \text{decrease} = & \frac{-1}{26} \times 100 = & \boxed{3\frac{11}{13}\%} \text{ Ans} \\ \% & & \end{array}$$

111 In a manufacturing unit, it was noted that the Price of raw material has increased by 25% and the labour cost has gone up from 30% of the cost of raw material to 38% of the cost of the raw material. what percentage of the consumption of raw material be reduced to keep the cost the same as that before the increase ?

[SSC CGL Pre 2022]

Soln:- $25\% \rightarrow \frac{1}{4}$

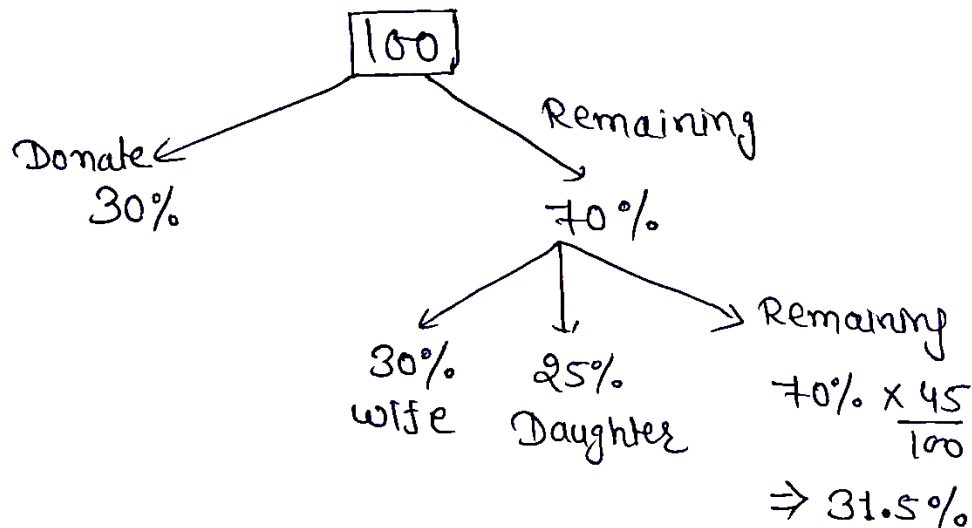
Price :- $\begin{array}{r} 40 \\ 50 \end{array} \rightarrow 38\%$
 Labour cost :- $\begin{array}{r} 12 \\ 19 \end{array}$

Consumption $\rightarrow 52 : 69$

$\frac{17}{69} \times 100 = \boxed{24.6\%}$ Ans

112 A man donates 30% of his wealth to charity. 30% and 25% of the remaining wealth to his wife and son respectively. The rest he divides equally between his three daughters. One of his three daughters gets Rs 42 lakh as her share. What was the man's wealth (in Rs lakhs)?

Soln:-



Each daughter's share = $\frac{31.5}{3} = \underline{10.5\%}$

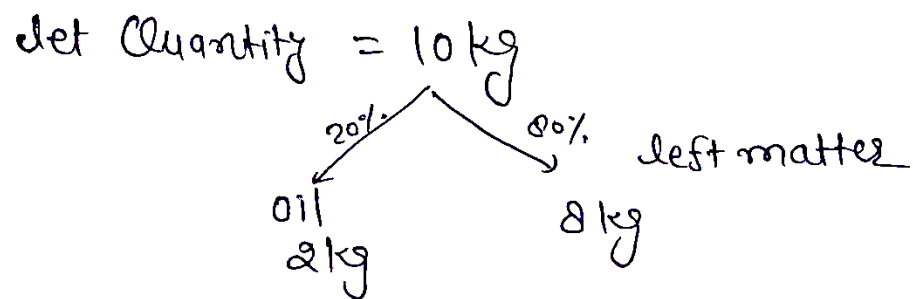
$10.5\% \rightarrow 42 \text{ lakh}$

$1\% \rightarrow 4 \text{ lakh}$

$100\% \rightarrow \boxed{400 \text{ lakh}}$ Ans

113 Oil equal to 20% of the weight of ground nut is extracted in a mill. The matter left after extraction is sold as cattle feed at the rate of Rs. 12.5/kg. The groundnuts are bought at Rs. 20/kg. The processing cost is Rs. 5/kg. At what price (Rs Per kg) should the oil be sold to earn 20% profit on total costs (Total cost = Cost of groundnuts and Processing costs)?

Soln :-



$$S.p = 2x + 8 \times 12.5 \quad (\text{let } x \text{ is oil S.p})$$

$$S.p = 2x + 100 \quad \text{--- (1)}$$

$$\begin{aligned} \text{Cost Price} &= 10 \text{ kg} \times 20 + \frac{5 \times 10}{\text{Processing}} \\ &= 250 \end{aligned}$$

$$\begin{aligned} 20\% \text{ profit then } S.p &= 250 \times \frac{120}{100} \\ &= 300 \quad \text{--- (2)} \end{aligned}$$

from (1) & (2)

$$2x + 100 = 300$$

$$2x = 200$$

$$\boxed{x = 100} \quad \underline{\underline{\text{Ans}}}$$

114 In an examination, the number of students who passed and the number of students who failed were in the ratio 25:4. If one more student had appeared and passed and the number of failed students was 3 less than earlier, the ratio of passed students to failed students would have become 22:3. What is the difference between the number of students who, initially, passed the examination and the number of students who failed the examination?

Soln:- Basic:-

$$P : F = 25x : 4x$$

$$\text{Total} = 29x$$

$$\frac{\text{Fail}}{\text{Total}} = \frac{4x-3}{29x+1} = \frac{3}{25}$$

$$100x - 75 = 87x + 3$$

$$13x = 78$$

$$x = 6$$

$$\text{difference } 25x - 4x = 21x = 21 \times 6 = \boxed{126} \text{ Ans}$$

m-II Ratio method

$$21 \times 6 = \boxed{126} \text{ Ans}$$

$$\Rightarrow 12 + 66 = 78$$

$$(25 \times 3) - (22 \times 4) \rightarrow 78$$

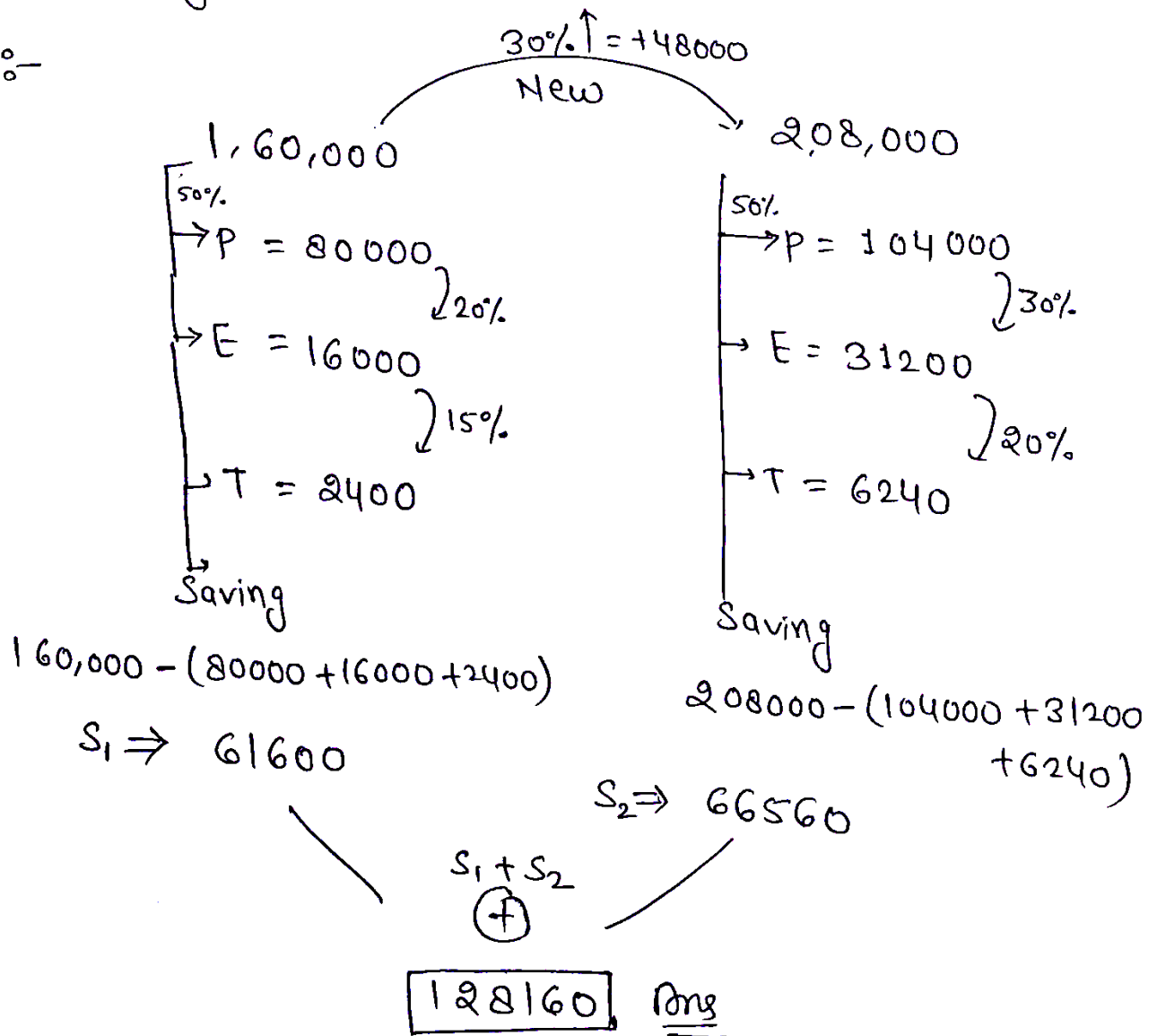
$$75 - 88 \rightarrow 78$$

$$13 \rightarrow 78$$

$$1 \rightarrow \textcircled{C}$$

115 The monthly salary of a person was Rs. 1,60,000. He used to spend on three heads - Personal and family expenses (P), Taxes (T) and Education loan (E). The rest were his savings. P was 50% of the income, E was 20% of P and T was 15% of E. When his salary got raised by 30%, he maintained the percentage level of P, but E became 30% of P and T became 20% of E. The sum of the two savings (in Rs.)

Soln:-



116] A man started off a business with a certain capital amount. In the first year, he earned 60% Profit and donated 50% of the total Capital (Initial amount + profit). He followed the same procedure with the remaining capital after the second and third year. If at the end of the three year, he is left with Rs. 15,360, what was the initial amount (in Rs.) with which the man started his business?

Soln:-

Let Initial amount = x

$$x \times \frac{160}{100} \times \frac{50}{100} \times \frac{160}{100} \times \frac{50}{100} \times \frac{160}{100} \times \frac{50}{100} = 15360$$

$$x \times \frac{16}{10} \times \frac{1}{2} \times \frac{16}{10} \times \frac{1}{2} \times \frac{16}{10} \times \frac{1}{2} = 15360$$

$$\boxed{x = 30,000} \text{ Ans}$$

117] A fruit seller sells 45% of the oranges that he has along with one more orange to a customer. He then sells 20% of the remaining oranges and 2 more oranges to a second customer. He then sells 30% of the now remaining oranges to a third customer and is still left with 5 oranges. How many oranges did the fruit seller have initially?

[SSC CGL - 2022]

Soln:- Let total number of oranges. = x

$$\left\{ \left(x \times \frac{55}{100} - 1 \right) \times \frac{80}{100} - 2 \right\} \times \frac{10}{100} = 5$$

$$\left(x \times \frac{55}{100} - 1 \right) \times \frac{80}{100} = 50 + 2$$

$$\left(x \times \frac{55}{100} - 1 \right) = \frac{52 \times 10}{8}$$

$$\frac{x \times 55}{100} - 1 = 65$$

$$\frac{x \times 55}{100} = 66$$

$$x = \frac{66 \times 100}{55}$$

$$\boxed{x = 120} \quad \underline{\text{Ans}}$$